GRADE 4

Computational Thinking

National Level Examination

SUBJECT CODE | | | |

Total Questions: 30 **Total Marks: 30** **Time: 45 Minutes**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **CAREFULLY CHECK YOUR NAME, CLASS, ROLL NUMBER, AND OTHER REQUIRED DETAILS ON THE OMR ANSWER SHEET.**
- **DO NOT FORGET TO SIGN THE OMR SHEET. ALSO WRITE YOUR ROLL NO. ON THE QUESTION PAPER BOOKLET.**
- **USE ONLY BLACK/BLUE BALL PEN AND HB PENCIL TO FILL THE OMR ANSWER SHEET.**
- **ALL QUESTIONS ARE COMPULSORY AND CARRY ONE MARK EACH.**
- **SECTION A: ABSTRACT & SPATIAL THINKING (6 QUESTIONS); SECTION B: PATTERN RECOGNITION (6 QUESTIONS); SECTION C: DECOMPOSITION, NUMBERS & DATA (6 QUESTIONS); SECTION D: ALGORITHMIC THINKING & LOGICAL SEQUENCING (6 QUESTIONS); SECTION E: OLYMPIAD CHALLENGE (6 QUESTIONS).**
- **ONLY OMR ANSWER SHEETS WILL BE EVALUATED.**
- **NO NEGATIVE MARKING FOR INCORRECT ANSWERS.**
- **YOU MAY USE THE SPACE PROVIDED ON THE LAST PAGE FOR ROUGH WORK (IF NEEDED).**
- **DIGITAL DEVICES LIKE CALCULATORS, SMART WATCHES, OR MOBILE PHONES ARE NOT ALLOWED.**
- **AFTER THE EXAM, SUBMIT BOTH THE OMR SHEET AND THE QUESTION PAPER TO THE INVIGILATOR.**

SAMPLE Paper

SECTION A – ABSTRACT & SPATIAL THINKING

Q1. A robot starts facing North. It moves 4 squares forward, turns right and moves 3 squares. Where is it from its starting point?
A. North-East
B. North-West
C. South-East
D. South-West

Q2. Which shape is seen when you look straight down at the top of a cone?
A. Triangle
B. Rectangle
C. Circle
D. Square

Q3. A square sheet is folded once vertically and once horizontally. One hole is punched away from the folds. How many holes will appear when the sheet is fully opened?
A. 2
B. 3
C. 4
D. 8

Q4. Three identical cubes are joined in a straight row, face to face. How many faces are exposed altogether?
A. 12
B. 14
C. 16
D. 18

Q5. Meera is facing West. She makes one quarter-turn to her left. Which direction is she facing now?
A. North
B. South
C. East
D. West

Q6. A square looks exactly the same after a quarter-turn. How many quarter-turns make one complete turn?
A. 2
B. 3
C. 4
D. 6

SECTION B – PATTERN RECOGNITION

Q7. Find the next number: 6, 11, 16, 21, ___
A. 24
B. 25
C. 26
D. 27

Q8. Find the missing number: 2, 6, 12, 20, 30, ___
A. 36
B. 40
C. 42
D. 44

Q9. Which letter comes next? A, D, H, M, ___
A. Q
B. R
C. S
D. T

Q10. The pattern repeats: ●, ■, ▲, ◆, ●, ■, ▲, ◆ ... Which symbol will be at the 18th position?
A. ●
B. ■
C. ▲
D. ◆

Q11. Find the next number: 3, 6, 12, 24, ___
A. 36
B. 42
C. 48
D. 54

Q12. Find the next number: 1, 4, 10, 19, 31, ___
A. 40
B. 43
C. 46
D. 49

SECTION C – DECOMPOSITION, NUMBERS & DATA

Q13. I am a 4-digit number. My thousands digit is 4. My hundreds digit is half of 4. My tens digit is 3 more than my hundreds digit, and my ones digit is 1. What number am I?
A. 4251
B. 4521
C. 4231
D. 2451

Q14. Four boxes contain 6 pencils each. If 5 pencils are used, how many pencils are left?
A. 17
B. 18
C. 19
D. 20

Q15. A class chart shows: Red - 8, Blue - 11, Green - 7, Yellow - 9. How many more students chose Blue than Green?
A. 2
B. 3
C. 4
D. 5

Q16. Rohan has ₹100. He buys a book for ₹37 and a pen for ₹28. How much money is left?
A. ₹25
B. ₹30
C. ₹35
D. ₹45

Q17. School starts at 8:15 a.m. Assembly lasts 25 minutes and the first class lasts 40 minutes. At what time does the first class end?
A. 9:10 a.m.
B. 9:15 a.m.
C. 9:20 a.m.
D. 9:25 a.m.

Q18. There are 36 stickers packed equally into packets of 6. Aman gives away 2 complete packets. How many stickers are left?
A. 12
B. 18
C. 24
D. 30

SAMPLE Sample Paper

SECTION D – ALGORITHMIC THINKING & LOGICAL SEQUENCING

Q19. A number machine follows two steps: Add 4, then multiply by 2. What is the output for input 7?
A. 18
B. 20
C. 22
D. 24

Q20. A machine multiplies a number by 3 and then subtracts 2. The output is 25. What was the input?
A. 7
B. 8
C. 9
D. 10

Q21. Start with 2. Repeat these steps twice: Add 3, then multiply by 2. What is the final number?
A. 20
B. 22
C. 24
D. 26

Q22. A robot moves 3 squares East, turns left and moves 2 squares, then turns left and moves 1 square. Where is it from its starting point?
A. North-East
B. North-West
C. South-East
D. South-West

Q23. A robot is meant to draw a square using: Move 3 steps, Turn Right, repeated four times. One Turn Right is accidentally written as Turn Left. What should be corrected?
A. Change the wrong Turn Left to Turn Right
B. Remove one Move command
C. Add 3 more steps
D. Turn Left every time

Q24. These actions follow a rule: Write your name before covering the notebook; cover the notebook before placing it in the bag. Which action must happen second?
A. Place it in the bag
B. Cover the notebook
C. Write your name
D. Any action can be second

SECTION E – OLYMPIAD CHALLENGE

Q25. I am a 3-digit number with different digits. My hundreds digit is twice my ones digit. My tens digit is one less than my hundreds digit. The sum of my digits is 14. What number am I?
A. 432
B. 653
C. 764
D. 874

Q26. Find the next number: 1, 3, 2, 6, 5, 15, 14, ___
A. 28
B. 30
C. 42
D. 45

Q27. Four children A, B, C and D are of different ages. A is older than B. C is younger than B. D is older than A. Who is the second oldest?
A. A
B. B
C. C
D. D

Q28. Two stars together have a value of 14. One triangle has a value 3 less than one star. What is the value of one star plus one triangle?
A. 9
B. 10
C. 11
D. 12

Q29. Use this rule: If the number is even, divide it by 2. If it is odd, add 5. Start with 14 and apply the rule four times. What is the final number?
A. 3
B. 5
C. 6
D. 7

Q30. A robot must reach a point using exactly 2 moves Right and 2 moves Up. It may make these four moves in any order. How many different shortest routes are possible?
A. 4
B. 5
C. 6
D. 8

SAMPLE Sample Paper